

The empirical recurrence for OEIS A295780: recovering a conjecture that is not written down

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Abstract

OEIS A295780 records “Empirical recurrence of order 91”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 337$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 91, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 91 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A295780 is “Number of $n \times 7$ 0..1 arrays with each 1 adjacent to 0 or 2 king-move neighboring 1s.”. It has offset 1 and begins

34, 739, 7479, 127731, 1905471, 27085621, 409525082, 6037613762, ...

The entry states:

Empirical recurrence of order 91 (see link above)

As of the “Last modified” line on the live entry (Nov 27 2017 07:14:54, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 16512$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 337$ of the 16512, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 337$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 674$ exact terms of the model returns order 91 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{91} c_i a(n-i),$$

c_1	20	c_{24}	−56793719746069	c_{47}	−8352659120824270263	c_{70}	−6192064889008359
c_2	−18	c_{25}	−345485818016852	c_{48}	14919325479987396583	c_{71}	2192335432305554
c_3	−485	c_{26}	688227905733856	c_{49}	−4002175523766105735	c_{72}	700907072927078
c_4	−10421	c_{27}	281710467845201	c_{50}	−12804662012589817601	c_{73}	−1589584243156363
c_5	49399	c_{28}	−1902007271593479	c_{51}	13904818423410110390	c_{74}	−274362124586085
c_6	320944	c_{29}	6698766004030346	c_{52}	107818209291923339	c_{75}	−30562608512405
c_7	410960	c_{30}	1158132295587538	c_{53}	−12961954475616177326	c_{76}	−41914810652612
c_8	−10396963	c_{31}	−9382153038835610	c_{54}	7399095715801570683	c_{77}	64642628423
c_9	−29030219	c_{32}	15669828974163225	c_{55}	4100156489007653495	c_{78}	19116556693588
c_{10}	109329494	c_{33}	−7627309457452304	c_{56}	−6586801098577226447	c_{79}	7557226235073
c_{11}	395703573	c_{34}	−66354351087714035	c_{57}	1983055783605941156	c_{80}	−2818150837518
c_{12}	775018154	c_{35}	4792707380268162	c_{58}	3241302098158827124	c_{81}	−1292461040880
c_{13}	−3389264735	c_{36}	135084505275480127	c_{59}	−1810245006052080480	c_{82}	−181391187594
c_{14}	−9216537746	c_{37}	−388551516036451377	c_{60}	−184924564132350096	c_{83}	−14663928904
c_{15}	−3746754125	c_{38}	−100181475754577440	c_{61}	905318283061817531	c_{84}	21477379988
c_{16}	11219980736	c_{39}	1479482086550323187	c_{62}	−323484771507447410	c_{85}	1939714899
c_{17}	−20995185676	c_{40}	−1438302237326985671	c_{63}	−192476685464995456	c_{86}	−657460230
c_{18}	−166755258172	c_{41}	−1249271461217415673	c_{64}	64274803510712382	c_{87}	29411158
c_{19}	2450356299626	c_{42}	5095177540699525444	c_{65}	−18580041373429272	c_{88}	−163298
c_{20}	908750218841	c_{43}	−3371628751867856593	c_{66}	−11422183115438520	c_{89}	15037
c_{21}	−5092282108731	c_{44}	−4646957043225760099	c_{67}	−4676856181691514	c_{90}	−392
c_{22}	−5585007800762	c_{45}	10081417060293512169	c_{68}	1790542448316825	c_{91}	4
c_{23}	5562921301414	c_{46}	−4755884234996972599	c_{69}	4396217450914613		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 91, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $91 + 4$ terms removed returns order 91 as well, so $\deg q_{\min} = 91 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $91 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 91 that this sequence satisfies — and that family turns out to have exactly one member.

References

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